

3DXpert Monitor

Installation & Setup Guide

This guide walks you through installing 3DXpert Monitor on any Windows machine. The tool is a lightweight local web application that monitors 3DXpert.exe resource usage (Commit memory, GDI Objects, GPU memory) and tails the application log in real time.

Developed for Oqton QA. Node.js and NVIDIA drivers are the only external prerequisites.

System Requirements

Component	Requirement	Where to get it
Windows	Windows 10 / 11 (64-bit)	Pre-installed
Node.js	v18 LTS or later	nodejs.org/en/download — choose the Windows Installer
npm	Included with Node.js	Installed automatically with Node.js
NVIDIA driver	Any recent driver (for GPU compatibility)	nvidia.com/drivers — optional, column shows n/a with CUDA
3DXpert	18.0 (installed at default path)	Pre-installed by Oqton
Web browser	Chrome, Edge, or Firefox	Pre-installed on Windows

What Is Included in the Package

- dist/ — pre-built application (server + frontend). No build step needed.
- node_modules/ — all Node.js dependencies (offline-ready after install).
- install.bat — automated installer. Run this first.
- run_launcher.bat — one-click launcher. Run this after installing.
- server/, client/ — source code (optional, for reference or modification).

Quick Start

- 1** Install Node.js
Download and run the Windows Installer from nodejs.org. Accept all defaults. Restart your machine if prompted.
- 2** Copy the package folder
Copy the 3dxdperts-launcher folder to any location on the target machine, e.g.
C:\QA_Stuff\Tools\PS_Scripts\3dxdperts-launcher\

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- 3** Run install.bat
Double-click install.bat inside the folder. It will verify Node.js, install npm dependencies, and check for NVIDIA drivers. A success message is shown when complete.
 - 4** Launch the app
Double-click run_launcher.bat. The server starts and the app opens in your default browser at <http://127.0.0.1:3000>.
 - 5** Start monitoring
Click Start in the sidebar. The Resources panel updates on every refresh interval, and the Log Reader tails the latest 3DXpert log file.
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Step-by-Step Installation

Step 1 — Install Node.js

Visit <https://nodejs.org/en/download> and download the Windows Installer (.msi) for the LTS version (v18 or later).

- Run the installer and click through with default settings.
- Make sure the option "Add to PATH" is checked (it is by default).
- Restart the machine if Windows asks you to.
- Verify: open a Command Prompt and run `node --version`. You should see v18.x.x or higher.

Step 2 — Copy the Package

Copy the entire 3dexpert-launcher folder to the target machine. A USB drive or network share both work. The recommended location is:

```
C:\QA_Stuff\Tools\PS_Scripts\3dexpert-launcher\
```

Any writable folder is fine. Avoid paths with special characters or very long names.

Step 3 — Run install.bat

Double-click install.bat inside the 3dexpert-launcher folder. A console window will open and perform the following checks automatically:

- Confirms Node.js is installed and readable from PATH.
- Runs npm install to install all JavaScript dependencies from the included package.json.
- Checks whether nvidia-smi is available (non-fatal — GPU column shows n/a if absent).

If npm install fails due to a network error, ensure the machine has internet access, or copy the node_modules folder directly from a machine where install.bat already succeeded.

Step 4 — Launch

Double-click run_launcher.bat. This script:

- Opens a hidden PowerShell window.
- Sets the port to 3000 and starts the Node.js server.
- Opens your default browser at `http://127.0.0.1:3000`.

Do not close the PowerShell window that appears in the taskbar — it hosts the server. Use the Exit button inside the app to shut everything down cleanly.

Using the App

Sidebar Controls

All monitoring controls are in the left sidebar inside the Monitor card:

- 3DXpert version — shows the FileVersion of 3DXpert.exe, read at startup.
- Refresh interval — how often the Resources panel polls (2 / 5 / 10 / 30 seconds). Default: 5 s.
- Auto-stop after — monitoring stops automatically after this many minutes (2 / 5 / 10 / 30). Default: 5 min.
- Max rows shown — maximum number of rows in both panels (10 / 50 / 100 / 200). Default: 50.
- Start / Stop — begins or halts polling.
- Clear — erases collected data (visible only when stopped).

Resources Panel (right)

Shows a live time-series table with one row per refresh interval:

- Time — timestamp in HH:MM:SS (24 h). Green dot = CPU active.
- Commit — Commit Size from Windows (kB), mapped to Task Manager's Commit column.
- GDI Objects — GDI handle count for 3DXpert.exe.
- GPU Mem — VRAM used by the NVIDIA GPU (kB), read via nvidia-smi.

Below the table, trend sparkline charts plot the last 60 samples for Commit, GDI, and GPU Mem. Each series scales independently so small changes are visible.

Log Reader Panel (left)

Tails the most recently modified 3DXpert*.log file from:

```
C:\ProgramData\Oqton\3DXpert\18.0\Data\Log\
```

The file is copied to a temporary location before reading to avoid locking the active log. Date prefixes are stripped — lines start with the time value only.

Folder Shortcuts

The Open folder group in the sidebar opens Windows Explorer directly at:

- DUMP — C:\Windows\Temp_3DXpertDumps
- DATA — C:\ProgramData\Oqton\3DXpert\18.0\Data
- PROGRAM — C:\Program Files\Oqton\3DXpert\18.0\Program

Exit

Click the red Exit button at the bottom of the sidebar. This sends a shutdown signal to the server, closes the PowerShell host window, and closes the browser tab.

Troubleshooting

Browser does not open

Navigate manually to <http://127.0.0.1:3000> after double-clicking `run_launcher.bat`. Wait 3–5 seconds for the server to start.

Port 3000 already in use

Another application is using port 3000. Edit `run_launcher.bat` and change the line:

```
set PORT=3000
```

to a different port number, e.g. 3001. Reload the browser at the new port.

GPU column shows n/a

`nvidia-smi` is not found in PATH. Install or update the NVIDIA display driver from [nvidia.com/drivers](https://www.nvidia.com/drivers). After installing, close and re-open `run_launcher.bat`.

GDI Objects shows 0

`3DXpert.exe` is not running, or the process handle could not be opened. Ensure 3DXpert is running and the app is not sandboxed by group policy.

Log Reader is empty

Confirm the log directory exists:

```
C:\ProgramData\Oqton\3DXpert\18.0\Data\Log\
```

And that at least one file matching `3DXpert*.log` exists inside it.

install.bat fails at npm install

Check internet connectivity. If the machine is air-gapped, copy the `node_modules` folder from a machine where `install.bat` already succeeded (same Node.js version required).

PowerShell execution policy error

Run the following in an Administrator PowerShell, then retry:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Third-Party Components

The following open-source packages are used by 3DXpert Monitor. All are distributed under permissive licenses compatible with internal enterprise use.

Component	Requirement	Where to get it
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Node.js	MIT / custom	nodejs.org — JavaScript runtime for the server
Express	MIT	expressjs.com — HTTP server framework
React	MIT	react.dev — UI library
Vite	MIT	vitejs.dev — Frontend build tool
Tailwind CSS	MIT	tailwindcss.com — Utility-first CSS framework
shadcn/ui	MIT	ui.shadcn.com — Accessible UI component library
Lucide React	ISC	lucide.dev — Icon library
TanStack Query	MIT	tanstack.com/query — Data-fetching and caching
Wouter	MIT	github.com/molefrog/wouter — Lightweight client router
TypeScript	Apache-2.0	typescriptlang.org — Typed JavaScript
NVIDIA nvidia-smi	Proprietary (NVIDIA)	Included with NVIDIA display driver. Used to query GPU

No telemetry, analytics, or external network connections are made by the app at runtime. All data stays on the local machine.